Jun-Kun Chen

Ph.D. Candidate in Computer Science | Expected Dec. 2026

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Legal name: Junkun Chen | Publications: Jun-Kun Chen / J.-K. Chen

Research Focus

Research in **diffusion-based controllable generation**, with first/co-first-author work at NeurIPS and CVPR and research internships at Meta and SpreeAI. Expertise spans **multimodal conditioning**, temporally consistent image/video generation, and visual world models, with current work on multimodal post-training.

Industry Experience

Meta Reality Labs — **May–Dec. 2025; May–Aug. 2023**
Research Scientist Intern (2025 continued part-time Aug.–Dec.) — Bay Area, CA, U.S. / Zurich, Switzerland

- **2025:** Designed parallel training workflows for 3D/video generation at **128–256-GPU scale** across multi-node clusters.
- **2023:** Developed multi-view conditioning, structured noise, and self-supervised consistency training for diffusion editing; published as *ConsistDreamer* at CVPR 2024.

SpreeAI — **May 2024 – May 2025**
AI Research Intern — Remote, U.S.

- **Dress&Dance**: introduced **multimodal conditioning** and multi-stage training for high-fidelity, controllable try-on video generation.
- **Virtual Fitting Room** (NeurIPS 2025): extended this work to **minute-scale, identity-consistent generation** with anchored autoregression; evaluated up to 90 seconds using short-clip training.

Current Research: Generative World Models

Research direction: connecting language reasoning and visual generation through training grounded in predicted outcomes.

Computer Use — Action-Conditioned Image-Diffusion World Model

- Developed **diffusion post-training** for a SenseNova MoT model predicting the next UI screenshot from a screen and action.
- Combined **structured consequence supervision** with agent-interaction data for separate Reasoner and Generator training.

Physics — Video-Diffusion World Model for Physical Events

- Adapted **Cosmos3 video diffusion** for physical-event prediction from observed RGB history.
- Implemented **online teacher distillation**, training the language Reasoner and diffusion Generator from video prediction errors.

Robotics — Motion-Conditioned Video Diffusion World Model

- Built an initial synthetic **video-diffusion prototype** for robot–object interaction prediction from a scene and prescribed motion.
- Designed **visual-proxy and paired-correspondence conditioning** to represent motion across robot–world setups.

Selected Publications

1. Virtual Fitting Room: Generating Arbitrarily Long Videos of Virtual Try-On from a Single Image ↗ — **NeurIPS 2025**
   Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Sep. 2025)
2. ProEdit: Simple Progression is All You Need for High-Quality 3D Scene Editing ↗ — **NeurIPS 2024**
   Jun-Kun Chen, Yu-Xiong Wang — Project | arXiv (Nov. 2024)
3. ConsistDreamer: 3D-Consistent 2D Diffusion for High-Fidelity Scene Editing ↗ — **CVPR 2024**
   Jun-Kun Chen, Samuel Rota Bulò, Norman Müller, Lorenzo Porzi, Peter Kontschieder, Yu-Xiong Wang — Project | arXiv (Jun. 2024)
4. Instruct 4D-to-4D: Editing 4D Scenes as Pseudo-3D Scenes Using 2D Diffusion ↗ — **CVPR 2024**
   Linzhan Mou*, **Jun-Kun Chen***, Yu-Xiong Wang — Project | arXiv (Jun. 2024)
5. HDEdit: Editing Videos and 3D Scenes with Video Diffusion Through Hierarchical Task Decomposition ↗ — **3DV 2026**
   Yanming Zhang*, **Jun-Kun Chen***, Jipeng Lyu, Yu-Xiong Wang — Project | Earlier version (arXiv, Mar. 2025)
6. Dress&Dance: Dress up and Dance as You Like It ↗ — **WACV 2027**, under review
   Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Aug. 2025)

*Equal contribution. Earlier publications also appear as Junkun Chen / J. Chen.

Education

University of Illinois Urbana-Champaign — **2021 – Dec. 2026 (expected)**
Ph.D. Candidate in Computer Science; Advisor: Prof. **Yu-Xiong Wang** — Urbana, IL

Tsinghua University — **2017 – 2021**
B.Eng. in Computer Science, Yao Class (IIIS); Major GPA: 3.94/4.0 — Beijing, China

Technical Skills & Honors

Tools & Methods: PyTorch; Python; C++; image/video diffusion; Transformers; vision-language models; multimodal conditioning; data curation; benchmark design; quantitative and human evaluation; diffusion post-training.

Competition Highlights: ICPC EC-Final Gold (4th); ICPC Qingdao Gold (2nd); CCPC Harbin Gold; NOI Gold.

Selected Research Contributions

Controllable Try-On to Long-Horizon Video — Dress&Dance and VFR

- **Dress&Dance:** developed **multimodal conditioning**, paired-triplet data, and multi-stage training; achieved **20–59% lower FVD** than a Kling 1.6 try-on-and-animation pipeline on Internet and captured evaluation sets.
- **Virtual Fitting Room:** extended short-clip training to anchored autoregressive generation; led VBench subject/background consistency and motion smoothness on 30–50 s hard-motion evaluations against FramePack and Kling 2.0.

General-Purpose Diffusion Editing — HDEdit and ProEdit

- Designed **hierarchical task decomposition** that separates instruction fulfillment from content preservation, combining LLM-based scheduling with interpolation-based refinement for training-free video and scene editing.
- Developed tuning-free preservation control and a **progressive editing workflow** with preview and edit-strength control, enabling high-fidelity geometry and appearance editing.

Consistent Scene and 4D Generation — ConsistDreamer and Instruct 4D-to-4D

- Created context-rich multi-view conditioning, **3D-consistent structured noise**, and self-supervised warping consistency for high-fidelity diffusion editing across views.
- Extended 2D diffusion to long-term dynamic scenes through **anchor-aware attention** and optical-flow-guided sliding windows.

Geometric Visual Representations — NeuralEditor and PointTree

- Built point-based radiance-field and relaxed k-d-tree representations for editable scene geometry and **transformation-robust point-cloud encoding**.

Additional Publications

7. SceneCraft: Layout-Guided 3D Scene Generation ↗ — **NeurIPS 2024**

8. Contrastive Learning Relies More on Spatial Inductive Bias Than Supervised Learning: An Empirical Study ↗ — **ICCV 2023**

9. NeuralEditor: Editing Neural Radiance Fields via Manipulating Point Clouds ↗ — **CVPR 2023**

10. PointTree: Transformation-Robust Point Cloud Encoder with Relaxed K-D Trees ↗ — **ECCV 2022**

11. Is Self-Supervised Contrastive Learning More Robust Than Supervised Learning? ↗ — **ICML 2022 Workshop**

12. TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery ↗ — **arXiv preprint, 2022**

13. RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs ↗ — **ICLR 2021**

14. Pre-training Assisted Scene-aware Dialogue Generation ↗ — **DSTC8 Workshop at AAAI 2020**

Earlier Experience

Mila – Quebec AI Institute — **Feb. – Aug. 2020**
Research Intern; knowledge-graph reasoning with Prof. Jian Tang — Montreal, Canada

Baidu, Natural Language Processing Department — **July 2019 – Jan. 2020**
Research / Development Intern; scene-aware dialogue generation (DSTC8 top-3 system) — Beijing, China

Competitions, Leadership & Coursework

Programming Competitions: ICPC EC-Final 2018, Gold Medal (4th); ICPC Qingdao Regional 2018, Gold Medal (2nd); CCPC Harbin Regional 2017, Gold Medal; NOI 2016, Gold Medal.
Additional Distinctions: CTSC 2017 (7th); CTSC 2016 Gold (8th); NOI Winter Camp 2016 Gold (3rd).
Leadership & Problem Setting: Problem setter for NOI, CTSC, ICPC, and CCPC events; Vice Chair, Tsinghua Student Association of Algorithms and Competitions (2018–2019).
Selected Coursework (Generative Models): Deep Generative Models (A+); Generative Models for Biomedicine and Life Sciences (A).
Vision & 3D: 3D Vision (A+); Deep Learning for Computer Vision (A); Computer Vision (A); Computational Photography (A+).
Learning: Meta-Learning (A+); Machine Learning (A).